

Industry Experiences

- **Process Engineer–Chagalesh Consulting Engineers Company Since Winter 2021**
 - **Basic Design** – Capacity Expansion, Iranian Chemical Industries (Esfahan, LAB):
 - Evaluated and upgraded utilities (water, steam, nitrogen, gas, and fuel oil) and designed auxiliary systems, including hot oil and cooling water services, for new units.
 - **Tender and Basic Design** – Utility units, Andimeshk Petrochemical
 - Designed and optimized utility units for wastewater treatment, including Electrodialysis Reversal (EDR), Reverse Osmosis (RO), and Zero Liquid Discharge (ZLD) systems, as well as cooling water treatment, steam condensate cycle, and air separation units.
 - **Detailed Design** – Designed Ethylene Oxide (EO) unit, Andimeshk Petrochemical
 - Working on equipment selection and integration with vendors such as centrifugal and reciprocating compressors, pumps, flare network design, and relief load calculations.
 - **Detailed Design** – CCR Platforming, NHT, BSU Units & Utilities, Tehran Refinery
 - Worked on PID development, instrument data sheets, cause-and-effect diagrams for CCR units, hydraulic calculations, and vendor reviews for steam turbines, air separation units, PSA systems, and centrifugal/reciprocating compressors; participated in HAZOP and SIL studies.
 - **Field Process Engineer** – BSU Unit, Tehran Refinery
 - Oversaw pre-commissioning, commissioning, and start-up activities, providing procedures for all stages; prepared early production and as-built drawings; responded to technical queries; and worked in close cooperation with operators at site.
 - **Feasibility study and Basic Design** – Off-site units, Hamedean Petrochemical
 - Designed off-site area for EO/EA/GE/ETX units, including storage tanks, loading and unloading systems for liquid, paste, and solid materials handling, drainage and isolation philosophy, and flare network design.
 - **Manager contractor** – Capacity Expansion, phase III of Ilam Gas Refinery
 - Worked on depressurization philosophy and calculations, flare network design, booster and transfer compressors, and amine treating and SRU unit simulation; participated in P&ID and 3D model review.
- **Software developer –In-house software developer, since Spring 2026**
 - Develop agent AI for automation of process engineering activities in EPC projects

- **Instructor– Chagalesh Consulting Engineers Company Since Winter 2021**
 - Relief load calculation and flare network design.
 - Reciprocating compressors design.
- **Process modeling engineer–R&D project, Amirkabir University of Technology, since Winter 2025**
 - Conducted wide-plant simulation for the air separation unit using Aspen HYSYS, integrating a dynamic process model for an operator training system (OTS).

Skills

- Prepared and analyzed various process flow diagrams (P&ID, PFD, UDD, UFD, BFD).
- Familiar with utility and offsite services, including air separation unit, cooling water circuits, hot oil rings, steam & water loops, fuel systems, wastewater treatment, storage tanks, and loading-unloading facilities.
- Sized and prepared datasheets for heat exchangers, heaters, air coolers, two-phase and three-phase separators, and towers.
- Conducted overpressure calculations, prepared Relief Load Summary documents, and performed sizing of flare networks and depressuring calculations.
- Performed process simulations in steady-state and dynamic modes.
- Performed hydraulic calculations using for steady-state and transient flow analysis.
- Familiar with rotary equipment such as pumps, compressors, steam turbines, and blowers.
- Familiar with instrumentation devices and control systems used in process industries.
- HAZOP, SIL and 3D model meeting participation.
- Control/shutdown/drainage/isolation/ philosophies, C&E and logic diagrams.
- Preparation and contribution to pre-commissioning, commissioning, and start-up procedures.
- Conducted vendor data review for equipment compliance, specification matching, and integration with process systems.
- Familiar with codes and standards such as TEMA, API, ASME, and design practices from companies including ExxonMobil, JGC, Foster Wheeler, GPSA, Chevron, Total, and Samsung.

Softwares

- **Simulator:** Aspen Hysys /Aspen energy analyzer/ Aspen Plus/ Aspen flarenet/ Wave
- **Piping simulation:** Olga/ PVT sim/ AFT Impulse
- **Programming:** Python/ Matlab/ GAMS
- **Economic analysis:** Aspen process economic analyzer/ Comfar
- **Equipment sizing:** HTRI/ Koch glitch/ Flaresim/ Aspen Adsorption
- **General:** Microsoft Office /Auto Cad/ Microsoft Visio

Education

Master of Science in Chemical Engineering

2021 – 2024

Amirkabir University of Technology (Tehran Polytechnic)

- Overall GPA: 4/4 (18.56/20)

Bachelor of Science in Chemical Engineering

2017 – 2021

Amirkabir University of Technology (Tehran Polytechnic)

- Overall GPA: 3.78/4.00 (17.73/20)

Language Proficiency

- **ENGLISH:** Full professional proficiency